

Patient Profile

Name: Yash M. Patel

Age: 21 Years

Gender: Male

Summary of Results

Serum liver function panel shows an isolated elevation of ALT (100.5 U/L) while AST remains within normal limits (16 U/L). The AST:ALT ratio is 0.50, below the reference threshold of 1.00. Alkaline phosphatase is markedly low (15.4 U/L) and albumin is significantly decreased (2.0 g/dL). The direct bilirubin is low, and total bilirubin is within normal limits. The overall pattern suggests hepatic injury or metabolic liver dysfunction rather than cholestasis or significant hepatic synthesis impairment.

Detailed Test Explanations

- **AST (SGOT)** – 16 U/L (normal 15–40). The value is within reference limits; low or high AST indicates various causes of hepatocellular injury, but isolated normal AST does not exclude liver disease when ALT is elevated.
- **ALT (SGPT)** – 100.5 U/L (normal 10–49, *high*). ALT is more liver-specific; elevations reflect hepatocellular injury, commonly due to fatty liver, viral hepatitis, or medication toxicity.
- **AST:ALT Ratio** – 0.50 (<1.00). A ratio <1 indicates that ALT exceeds AST, typical for non-alcoholic fatty liver disease (NAFLD) and most viral hepatitis. Ratios >1 are often seen in alcoholic hepatitis or advanced fibrosis.
- **GGTP** – 10.2 U/L (normal 0–73). Within normal limits; this transaminase is sensitive to cholestasis and alcohol, supporting the lack of biliary obstruction.
- **Alkaline Phosphatase (ALP)** – 15.4 U/L (normal 30–120, *low*). A low ALP may reflect nutritional deficiency, hypophosphatasia, or a transient laboratory artifact; it is not typical of liver disease but can coexist with other abnormalities.
- **Bilirubin Total** – 0.60 mg/dL (normal 0.30–1.20). Within normal limits, indicating no significant impairment of bilirubin excretion.
- **Bilirubin Direct** – 0.10 mg/dL (<0.30). Normal.
- **Bilirubin Indirect** – 0.10 mg/dL (<1.10). Normal.
- **Total Protein** – 6.39 g/dL (normal 5.70–8.20). Normal, suggesting adequate protein synthesis.
- **Albumin** – 2.00 g/dL (normal 3.20–4.80, *low*). Albumin is a plasma protein synthesized by the liver; low levels may indicate chronic liver injury, malnutrition, or loss through kidneys.
- **A: G Ratio (Albumin/Globulin)** – 0.10 (normal 0.90–2.00). Indicates a disproportionately low albumin relative to globulin, consistent with hepatic synthetic dysfunction or protein loss.

Overall Interpretation

The pattern of isolated ALT elevation with a low AST:ALT ratio, combined with low albumin and a severely reduced A:G ratio, points toward hepatic injury likely secondary to non-alcoholic fatty liver disease (NAFLD) or early hepatocellular dysfunction. The lack of cholestatic markers (normal GGTP, bilirubin, ALP) and absence of hepatitis serology in the data suggest a non-viral etiology. Low albumin and the abnormal A:G ratio hint at early synthetic impairment, which warrants evaluation for possible fibrosis. Low alkaline phosphatase is atypical but can coexist with hepatic dysfunction; further confirmation may be required.

Recommendations

1. **Lifestyle Modification** – Adopt a balanced, low-sugar, low-fat diet; achieve a 5–10% weight loss if obese;

incorporate aerobic exercise 150 min weekly.

2. **Alcohol and Medication Review** – Avoid hepatotoxic drugs and limit alcohol intake to <2 drinks/week.
3. **Repeat Liver Function Tests (LFTs)** in 6–12 weeks to assess trend of ALT, AST, albumin, and A:G ratio.
4. **Non-Invasive Fibrosis Assessment** – Consider elastography (FibroScan) or serum fibrosis markers (like ELF) to evaluate for early fibrosis.
5. **Referral to Hepatology** if ALT remains $>2 \times$ upper limit, albumin falls further, or imaging suggests steatosis.
6. **Monitoring** – Annual screening for liver enzymes in young adults with metabolic risk factors.

Disclaimer

This report is for informational purposes only and should not replace professional medical advice. Please consult a qualified healthcare provider for personalized diagnosis and treatment.